Revision A Engineering Optimization and Final PCB Review

Perform a complete engineering review of this PCB and optimize it for a highly reliable Revision A prototype suitable for fabrication by JLCPCB.

The objective is **not** to minimize PCB size. The objective is to maximize electrical reliability, manufacturability, ease of assembly, ease of debugging, and long-term serviceability.

Do not remove or change functionality.

PCB Stackup

Convert the PCB to a four-layer stackup if it is not already.

Use:

Layer 1

- Components
- Primary signal routing
- Short local power traces

Layer 2

- Continuous, uninterrupted solid GND plane

Layer 3

- High-current protected 5V power plane
- High-current copper pours
- Short branches to loads

Layer 4

- Secondary signal routing
- Low-current routing only

Do not split the Layer 2 ground plane.

Do not route signals that unnecessarily interrupt the ground plane.

5V Power Distribution

This PCB may power:

- ESP32-S3
- PCA9685
- Two DFPlayer Mini modules
- OLED display
- Three PIR sensors
- Eight Derek ports
- Sixteen servos
- Addressable LEDs

The total available current is approximately 15A.

Redesign the power distribution using engineering best practices.

Requirements:

- Use Layer 3 as the primary 5V distribution plane.
- Eliminate narrow high-current bottlenecks.
- Use wide copper pours.
- Use wide trunk feeds from the protected input.
- Branch to groups of RJ45 connectors.
- Keep servo current away from logic circuitry whenever practical.
- Add multiple stitching vias wherever high-current power transitions between layers.
- Use generous copper around the fuse and reverse-polarity circuitry.

Grounding

Maintain a continuous Layer 2 ground plane.

Optimize return current paths.

Avoid unnecessary ground plane splits.

Provide additional ground stitching vias around:

- RJ45 connectors
- DFPlayer modules
- ESP32
- PCA9685
- OLED connector
- Board perimeter

Bulk Decoupling

Verify capacitor placement.

Minimum requirements:

- 4700µF bulk capacitor near power entry.
- 1000µF capacitor near the RJ45 connector bank.
- 100µF capacitor near every group of two Derek connectors.
- 0.1µF ceramic capacitor adjacent to every IC power pin.

Move capacitors closer if required.

High-Current Routing

Review every high-current path.

Increase copper width wherever practical.

Use copper pours instead of long thin traces.

Verify that no single trace becomes a current bottleneck.

ESP32 Placement

Verify:

- antenna is flush with PCB edge
- no copper exists beneath antenna
- no ground plane exists beneath antenna
- antenna keep-out area is maintained
- USB connector remains fully accessible
- sufficient cable clearance exists
- tall components are not located in antenna keep-out

DFPlayer Placement

Verify:

- both SD cards remain fully accessible
- sufficient finger clearance exists
- enclosure walls will not interfere

- audio routing remains separated from servo power routing

OLED

Verify:

- powered from 3.3V
- I2C routing is short
- connector remains accessible

PCA9685

Verify:

- module footprint matches the commercial Teyleten Robot PCA9685 module
- header spacing
- pin order
- mechanical clearance
- VCC = 3.3V
- Servo V+ = protected 5V

ESP32 Module

Verify:

- selected ESP32-S3 DevKit footprint
- pin spacing
- USB connector alignment
- antenna overhang
- module clearance
- header alignment

LED Driver

Verify:

- 74AHCT125 powered from protected 5V
- proper 3.3V-to-5V level shifting
- 330Ω series resistor
- unused channels terminated safely

Reverse Polarity Protection

Review the reverse-polarity protection circuit.

Optimize for approximately 15A continuous current.

If practical, replace Schottky diode protection with a P-channel MOSFET ideal-diode circuit.

PCB Placement

Improve placement to reduce routing complexity.

Reduce unnecessary vias.

Reduce trace crossings.

Increase spacing where possible.

Do not optimize for minimum PCB size.

Optimize for:

- clean routing
- manufacturability
- debugging
- hand assembly
- maintenance

Connectors

Verify:

- RJ45 spacing
- OLED connector access
- DFPlayer access
- USB cable access
- DIP switch access
- test header access
- mounting hole clearance

Silkscreen

Verify every connector is clearly labeled.

Add or improve labels for:

- power polarity
- NOT ETHERNET warning
- Cluster ID switch
- test headers
- revision number
- board name

Design Rule Review

Perform a complete engineering review.

Identify and automatically correct:

- DRC violations
- overlapping copper
- insufficient clearances
- floating inputs
- missing pull-ups
- missing pull-downs
- missing decoupling capacitors
- inadequate trace widths
- inaccessible connectors
- inaccessible SD cards
- inaccessible USB connector
- incorrect footprints
- silkscreen overlap
- copper islands
- isolated pours

Preserve all existing functionality while improving the design.

The resulting PCB should represent a professionally engineered Revision A prototype suitable for fabrication and hand assembly.